

AYUSH MADHUKAR

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EDUCATION

San José State University | *B.S. Computer Engineering, in progress*
Foothill College | *Associate degree studies, Computer Science*

January 2024 – Present
January 2022 – January 2024

SELECTED PROJECTS

Hermes | *Personal AI system*

Python · macOS · Tailscale · RAG

- Designed a persistent agent backend on an M4 Mac mini so desktop, laptop, and mobile clients share workflows and tools over a private mesh.
- Developing scoped memory and retrieval boundaries that surface useful context without exposing an unrestricted private notebook.

Obsidian RAG | *Knowledge retrieval*

Python · Embeddings · Retrieval

- Building a visibility-aware retrieval layer for grounded question answering across a structured Obsidian vault.
- Preserves note metadata and source context so retrieval remains inspectable as models and embedding backends change.

Ayumad.me | *Portfolio and public knowledge layer*

React · TypeScript · Vite · Vercel

- Built a responsive portfolio with dynamic project, system, gear, and Markdown article routes backed by a structured content layer.
- Created five content-aware visual modes with accessible controls, mode-specific transitions, and optimized canvas particle rendering.

Homelab | *Infrastructure*

Proxmox VE · ZFS · Linux · Virtualization

- Repurposed a ThinkStation P520 as a virtualization and storage host for durable services and isolated experiments.
- Organizes workloads around explicit recovery boundaries so experiments can change without destabilizing core services.

LEADERSHIP & EXPERIENCE

Data Science & AI Club | *Co-President*

Foothill College

- Integrated the Google Data Analytics curriculum into club programming and taught practical data manipulation and visualization workflows.
- Developed Owlbot, a GPT-3.5-based campus chatbot, and mentored members through a Random Forest project for the Kaggle Titanic competition.

OwlHacks | *Vice President of Logistics*

Foothill College

- Coordinated schedules, technical supplies, catering, and workshop logistics for a three-day technology event serving 150+ attendees.
- Led 30+ volunteers and committee vice presidents through event execution, including workshops covering game development, CAD, and React.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, HTML/CSS

AI & Data: PyTorch, TensorFlow, Pandas, retrieval-augmented generation, embeddings, agent workflows

Systems: Linux, Proxmox VE, ZFS, Docker, virtualization, Tailscale

Web & Tools: React, Vite, Vercel, Git, Fusion 360, SolidWorks